

Module 1: Introduction to Backend Development

1. What is Backend Development?

Backend development is the part of a web application that works behind the scenes. It handles data processing, business logic, and communication with databases.

Example

When you log in to Instagram:

1. You enter your username and password.
2. The backend checks the database.
3. If the details are correct, you are logged in.

Technologies

- Python
 - Django
 - PostgreSQL
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2. How Web Applications Work

A web application works through communication between the client, server, and database.

Process

1. User sends a request.
2. Server receives the request.
3. Server processes the request.
4. Database provides data if needed.
5. Server sends a response back.

Example

When you search for a video on YouTube:

- Browser sends a request.
- Server finds matching videos.
- Results are displayed on your screen.

Flow

Client → Server → Database

Client ← Server ← Database

3. Client

A client is a device or application that sends requests to a server and displays responses.

Examples

- Google Chrome
- Firefox
- Mobile Applications

Example

When you open Amazon in Chrome, Chrome acts as the client.

4. Server

A server is a computer or software that processes requests and sends responses.

Functions

- Run backend code
- Process requests

- Connect to databases

Example

When you search on Google, Google's servers process your request and return results.

5. Database

A database is a collection of organized data used to store and retrieve information.

Examples

- PostgreSQL
- MySQL
- SQLite

Example

Instagram stores user profiles, passwords, and posts in a database.

Benefits

- Organized storage
 - Fast retrieval
 - Security
-

6. API (Application Programming Interface)

An API allows different applications to communicate with each other.

Example

A weather app uses a weather API to fetch weather information from a weather server.

Benefits

- Data sharing
 - Easy integration
 - Faster development
-

7. Frontend vs Backend

Frontend

The part of an application visible to users.

Technologies

- HTML
- CSS
- JavaScript

Example

Buttons, forms, images, and menus on a website.

Backend

The hidden part that processes data and handles logic.

Technologies

- Python
- Django
- Java

Example

Checking login credentials and saving user information.

8. What is Python?

Python is a high-level programming language known for its simplicity and readability.

Features

- Easy to learn
- Simple syntax
- Large community support

Example

```
name = "Akhil"  
print(name)
```

Uses

- Web Development
 - Data Analytics
 - Artificial Intelligence
 - Automation
-

9. Why Python is Used in Enterprise Applications

Many companies use Python because it increases development speed and productivity.

Advantages

- Easy to write and understand
- Large number of libraries
- Fast development
- Scalable applications

Example

Companies such as Google, Instagram, and Netflix use Python in various systems.

10. Introduction to Django Framework

Django is a Python framework used to build web applications quickly and securely.

Features

- Admin Panel
- Authentication System
- ORM
- Security Features

Example

An online shopping website can be developed using Django.

Benefits

- Faster development
 - Less code
 - Easy database integration
-

11. Introduction to PostgreSQL Database

PostgreSQL is an open-source relational database management system.

Features

- Reliable
- Secure
- High Performance

Example

A company can store employee records in PostgreSQL.

Stored Data

- Employee Information
 - Customer Records
 - Product Details
-

12. Software Development Life Cycle (SDLC)

SDLC is a process used to develop software in a structured manner.

Stages

1. Requirement Gathering
2. Planning
3. Design
4. Development
5. Testing
6. Deployment
7. Maintenance

Example

If a company wants a student management system:

- Requirements are collected.
- Design is prepared.
- Developers write code.
- Testing is performed.
- Software is released.

13. Agile Methodology

Agile is a software development methodology based on continuous improvement and teamwork.

Key Concepts

- Small releases
- Continuous feedback
- Team collaboration

Example

Instead of building a complete application in six months, developers release features every two weeks.

Benefits

- Faster delivery
- Better quality
- Easy changes

14. Jira Workflow

Jira is a project management tool used to track tasks and project progress.

Uses

- Task Management
- Sprint Planning
- Bug Tracking

Workflow

To Do → In Progress → Testing → Done

Example

Task: Create Login Page

- To Do → Task assigned
 - In Progress → Developer working
 - Testing → QA testing
 - Done → Task completed
-

Conclusion

Backend development is responsible for handling application logic, data processing, and database communication. Python, Django, and PostgreSQL are widely used technologies for backend development. Understanding SDLC, Agile, and Jira helps developers work effectively in professional software development environments.